

Sri Lanka Institute of Information Technology



IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 07

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Question 1:

- (i) How many new processes are created in the program? Why?
 - 10 new processes
 - Each fork() creates 1 new child process
- (ii) Which process, the parent or the child, executes function _A()? Why?
 - Child process executes function _A(), because pid == 0 → child process
- (iii) Whose PID, the parent or the child, is printed in Line A? Why?
 - Parent prints the CHILD's PID
- (iv) What is the purpose of the loop (with its wait()) in Line B?
 - To wait for all child processes to finish execution

Question 2:

- (i) For N=5, How many processes are created when the program is executed?
 - 31 processes
- (ii) Modify the program so that only the parent process creates 3 child processes, and each new created process calls a function CPU(). In addition, make the parent process wait for each child's termination.

```
#include <stdio.h>
#include <unistd.h>

int main() {
    int i;
    pid_t pid;

    for(i=0; i<3; i++){
        pid=fork();

        if(pid == 0) {
            CPU();
            return 0;
        }
    }

    for(i = 0; i < 3; i++){
        wait(NULL);
    }

    return 0;
}
```

Question 3:

What will be the output in Line A?

- PARENT: value = 40

Question 4:

- (i) How many times will the fork () function be called in Program A? (i.e., how many processes are created?) Justify your answer.
 - 16 processes (including parent)
- (ii) What is the output of Line A in Program B? Justify your answer.
 - Value = 45
 - Value = 15

Question 5: Threads

1. What is the role of pthread_t tids[NUM_THREADS]?
 - Stores IDs of all created threads
2. What is the purpose of the for loop that calls pthread_create()?
 - To create multiple threads concurrently
3. What is the role of the function thread_routine()?
 - Defines the task each thread performs
4. Why is pthread_join() used in the second for loop?
 - To wait for all threads to complete execution